

VISHESH RANKA

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Seeking Supply Chain / Operations internship for Spring 2026 | Open to full-time roles from May 2027

PROFESSIONAL SUMMARY

Supply chain and operations professional with hands-on experience across procurement, MRO operations, and production engineering, backed by demonstrated analytics capability in demand forecasting, multi-echelon inventory optimization, and simulation using SQL, Python, and linear programming. Track record of cutting turnaround times, eliminating stockouts, and improving cross functional efficiency through data driven decision making. Pursuing an M.S. in Engineering Management and a Graduate Certificate in Optimization and Supply Chain Management at USC Marshall, applying quantitative rigor to supply chain management and analyst roles.

SKILLS

Supply Chain Management: Demand Forecasting, Inventory Management (EOQ, Safety Stock, Reorder Point), Multi-Echelon Inventory Optimization, Procurement & Sourcing, Supplier Evaluation & Vendor Management, MRO Operations, Production Planning, Lean Manufacturing, Supply Chain Optimization, Logistics Management, Process Improvement, Root Cause Analysis, Quality Control (ISO 9000), ERP (SAP)

Data & Analytics: SQL (DuckDB), Python (Pandas, LightGBM, PuLP), Linear Programming, Monte Carlo Simulation, Data Visualization (Plotly, Power BI), Microsoft Excel (Pivot Tables, VLOOKUP, Solver)

Engineering Software: MATLAB, AutoCAD, SolidWorks

EXPERIENCE

NTT Data Limited

December 2024 – February 2025

Procurement & Business Operations Intern

Mumbai, India

- Enabled faster, data backed purchasing decisions by building structured Excel vendor scorecards, benchmarking 4–6 suppliers per sourcing cycle on price, lead time, technical specs, and reliability
- Sustained cost-effective, on-time-in-full (OTIF) procurement across a large scale technology supply chain by coordinating 2–3 times weekly with 4–6 vendors on real time quotes, lead times, and delivery status

Air India Engineering Services Limited

June 2024 – October 2024

Mechanical Engineering Intern

Mumbai, India

- Reduced aircraft servicing turnaround time by 10% through root-cause analysis and testing of hydraulic, pneumatic, and avionics components, supported by precision equipment calibration and enhanced diagnostics
- Cut spare-parts stockouts by 5% by designing and implementing an automated inventory tracking system for critical avionics parts, optimizing reorder points and improving supply chain visibility

George Philips Medical Engineering Private Limited

January 2023 – April 2023

Production Engineering Intern

Mumbai, India

- Reduced production cycle time ~12% by redesigning the syringe pump production workflow and automating quality checks on a 500–2,000 units/day line, applying lean process improvement in cross functional collaboration with QA and manufacturing
- Improved material utilization by 12% through root cause analysis of production datasets covering cycle times, defect logs, and material usage for the HDM-DPT kit

PROJECTS

Hierarchical Demand Forecasting & Reconciliation

DuckDB (SQL), Python (LightGBM, Pandas), Nixtla hierarchical forecast

- Built an end-to-end forecasting pipeline for 30,490 product-store combinations from a 58M row Walmart (M5) dataset staging raw data in SQL and generating LightGBM demand forecasts across 5 business levels (company, category, department, store, item)
- Diagnosed a crossed (non-nested) hierarchy structure where product and geography rollups intersect rather than nest, and applied MinT reconciliation to enforce forecast consistency across all levels
- Improved company-wide forecast accuracy by 12.5% and achieved exact coherence (0% mismatch) across all hierarchy levels, versus up to 2.5% inconsistency in unreconciled forecasts

Multi-Echelon Inventory Optimization Under Demand Uncertainty

Python (PuLP, NumPy, Pandas), Linear Programming, Monte Carlo Simulation, Plotly, Streamlit

- Built a multi-echelon inventory optimization model (linear programming, PuLP) across a 13 node, 3 region global supply network, allocating safety stock between distribution centers and stores under a 95% service-level constraint
- Diagnosed autocorrelated demand causing standard safety stock formulas to understate distribution center variability by up to 10.6x over 30-45 day international lead times, and replaced it with empirical quantile-based sizing
- Validated via out of sample, rolling-window Monte Carlo simulation across 4 quarters, eliminating distribution center stockouts (100% vs. 87.7% under the naive policy) and cutting holding cost up to 8.3% versus a decentralized baseline

EDUCATION

University of Southern California

Los Angeles, CA

Master of Science in Engineering Management

August 2025 – May 2027

Graduate Certificate in Optimization & Supply Chain Management

Mukesh Patel School of Technology Management & Engineering

Mumbai, India

Bachelor of Technology in Mechanical Engineering

2021– 2025